Sam: I had no idea how to do it.

Author: Yes, you can. Though I think you needn't. Let us present y_n in a different form: $\boxed{y_n = a_n n + b_n \frac{1}{n}}$

and demand that the coefficient a_n be equal to unity if n is odd, and to zero if n is even; the coefficient b_n should behave in quite an opposite manner. In this particular case these coefficients can be determined as follows:

$$\boxed{a_n = \frac{1}{2}\left[1-(-1)^n\right]; \; b_n = \frac{1}{2}\left[1+(-1)^n\right]}$$

Consequently, $y_n = \frac{n}{2}\left[1-(-1)^n\right] + \frac{1}{2n}\left[1+(-1)^n\right]$

Do in the same manner the other two examples.

Sam: Seq 9 $\rightarrow$ $y_n = \frac{1}{2n}\left[1-(-1)^n\right] - \frac{1}{2(n-1)}\left[1+(-1)^n\right]$ ✓ (Verified from Reader)

Seq 10 $\rightarrow$ n is odd: $\frac{1}{n}$ n is even $\rightarrow \frac{n}{n+1}$

$$\boxed{y_n = \frac{1}{2n}\left[1-(-1)^n\right] + \frac{n}{2(n+1)}\left[1+(-1)^n\right]}$$ ✓ (Verified from Reader)

Q) How to come up with the values of a_n and b_n? What is the general way to combine such sequences whose analytical expression for certain groups of positions is known?

Author: It is important to note that an analytical expression for the nth term of a given sequence is not necessarily a unique method of defining a sequence. A sequence can be defined, for example, by recursion (or the recurrence method) (Latin word recurrere means to run back). In this case, in order to define a sequence one should describe the first term (or the first several terms) of the sequence and a recurrence (or a recursion) relation, which is an expression for the nth term of the sequence via the preceding one (or several preceding terms).

Using the recurrence method, let us present sequence (1) as follows

$y_1 = 1$; $y_n = 2y_{n-1}$

Reader: It's clear. Sequence (2) can be apparently represented by formulas.

$y_1 = 5$; $y_n = y_{n-1} + 2$